MIL-C-60293A(MU)
18 June 1970
SUPERSEDING
MIL-C-60293(MU)
22 April 1966

MILITARY SPECIFICATION

CONTAINER, SHIPPING, REUSABLE: 10543526

1. SCOPE

1.1 This specification covers the fabrication and testing of one type reuable shipping container for Range Finder AN/GVS-3 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-E-529	Enamel, Alkyd, Semi-Gloss
TT-I-558	Ink, Marking Stencil, Opaque for Non-Porous Surfaces (Wood Boxes, Fiber Cartons, etc.)

Military

MIL-A-5092	Adhesive, Rubber Base, General Purpose
MIL-B-7883	Brazing of Steels, Copper, Copper Alloys, and Nickle Alloys
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture Sensitivity
MIL-W-8604	Welding of Aluminum Alloys, Process For
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-C-21567	Compound, Silicone, Soft Film
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, For Packaging
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Federal

FED-STD-595	Colors

FSC 1240

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3.3.1 Shell forming process.- Container shells shown on Figures 3 and 4 shall be thermo formed by any proven method normally associated with the plastic industry. Finished formed parts shall be entirely free of blisters, blemishes or thermo stress marks.

3.4 Fabrication.

3.4.1 Extrusion edge frames.- The upper extrusion edge (female), Drawing B10539157 and the lower extrusion edge (male), Drawing B10539158 shall be formed in one continuous operation by proven forming techniques and controlled tooling to provide uniformity of alignment, flatness and shall fit freely in each other when container is closed. The formed extrusions shall be free of chatter, tool marks and distortions. The extrusions shall be joined by welding or brazing in accordance with MIL-W-8604 or MIL-B-7883. Joints shall be ground smooth (see Figure 5).

3.4.1.1 Finish.- Prior to assembly to container shells, the extrusion edge frames shall be finished 7.1.2 (dyed black) of MIL-STD-171. All surfaces except cavities of extrusion shall be finished 21.3 of MIL-STD-171, using primer MIL-P-8585 Color "Y". The primer and top paint coat shall be baked. Finish paint color shall be Olive Drab, X24087 of FED-STD-595.

3.4.1.2 Bonding of edge frames.- The edge frames shall be attached to the container shells with adhesive conforming to Drawing B11730899 and primer conforming to Drawing A11730900, applied in such a manner so as to provide structural integrity and a pressure-tight joint. The edge frame shall be tool rolled and crimped around the periphery to assure positive attachment to container shells, (see Figure 1A).

3.4.2 Clamp assemblies.- Ten (10) clamp assemblies conforming to Part Number 10539217-2, shall be attached to lower edge frame as shown on Figure 1. Attachment shall be accomplished as shown on Figure 2. Touch up shall be accomplished with paint conforming to TT-E-529, Color X24087 of FED-STD-595.

3.4.3 Handles.- Two (2) handles, conforming to Part Number 10539216-2, shall be attached to lower edge frame, at each end of the container as shown on Figure 1. Attachment shall be accomplished as shown on Figure 2. Touch up shall be accomplished with paint conforming to TT-E-529, Color X24087 of FED-STD-595.

3.4.4 Packing.- Packing conforming to Part Number 8644392-1 shall be formed by vulcanizing or butt bonding. Butt bonding shall be accomplished with adhesive conforming to MIL-A-5092, Type II. Prior to inserting the packing in the female edge frame, adhesive conforming to MIL-A-5092, Type II shall be uniformly applied within the well. There shall be no evidence of excess adhesive in the packing well. The packing shall be surface treated with a soft film silicone compound conforming to MIL-C-21567.

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distortion, cracks, or delamination and shall remain pressure-tight.

3.8.2 Standard ambient temperature.- Subsequent to the requirements of 3.8.1, the container shall meet all performance requirements of this specification at a standard ambient temperature between +60° and +90°F.

3.8.3 Impact resistance (ambient temperatures).- The latched container assembly, fitted with a dummy load specified in 4.6.1, shall sustain two drops, once on the top and once on the bottom, from a height of 30 inches. Subsequent to the two drops, the container shall show no evidence of damage which will affect its intended function. The impacts shall not unlatch the cover from the body, and the container shall then meet the requirements of 3.8.6 and 3.7.

3.8.4 Impact resistance (low temperature).- The latched container assembly, fitted with dummy load, specified in 4.6.1, shall sustain two drops, once on the top and once on the bottom, from a height of 24 inches after being subjected to low temperature conditioning at -40°F. Subsequent to the two drops, the case shall show no evidence of damage which will affect its intended function. The impact shall not unlatch the cover from the body, and there shall be no evidence of cracks or fractures. The container assembly shall then meet the requirements of 3.8.6 and 3.7, after the container has returned to ambient temperature.

3.8.5 Extreme operating temperatures.- The container shall meet the performance requirements of this specification while exposed to, and thermally stabilized at ambient temperatures of -65°F and 150°F.

3.8.6 Pressure-tight.- The container shall be capable of maintaining a positive pressure of $1.^{+}_{-}\ ^{.1}_{.1}$ psi without loss of pressure for a period of 5 minutes.

3.9 Workmanship.- Containers fabricated to the requirements of this specification, shall be in accordance with proven commercial forming techniques and shall be free of defects that might affect strength, serviceability and protection of contents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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TABLE I - CLASSIFICATION OF DEFECTS

CLASS CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
101. Improper forming	3.3, 3.3.1	Visual, SME
102. Edge frames improperly formed welded, or brazed	3.4.1	Visual
103. Edge frames improperly finished	3.4.1.1	Visual
104. Improper bonding of edge frame, improper adhesive, edge frame not rolled or crimped	3.4.1.2	Visual
105. Improper clamps or assembly to containers	3.4.2, 3.7.1	Visual, 4.6.2.1
106. Improper handles or assembly to containers	3.4.3, 3.7.1	Visual, 4.6.2.1
107. Improper packing or improperly bonded to edge frame assembly	3.4.4	Visual
108. Improper card holders, card holders improperly attached to container	3.4.5	Visual
109. Improper relief valve, valve installed improperly	3.4.6	Visual
110. Incorrect decal	3.4.6.1	Visual
111. Cushion inserts improperly fabricated	3.5.1, 3.5.2	Visual
112. Cushion inserts improperly bonded within container	3.5.3, 3.7.3	Visual, 4.6.2.3
113. Marking incorrectly applied or missing	3.6	Visual
114. Cover and body	3.7.2	4.6.2.2
MINOR: AQL 4.0 Defective		
201. Excess adhesive on gasket	3.5.4	Visual

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- Two sample containers shall be selected at random as a special sampling from each lot produced. The samples shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirements and tests.

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4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.2.1 Preparation for tests.- The container assembly shall be fitted with a dummy load constructed as shown on Figure 8. The pressure relief valve shall be removed and the aperture sealed with a gasketed fitting when performing the pressure-tight test.

4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Storage.- The container assembly with dummy load shall be tested as follows:

(a) Elevated temperature.- The container assembly shall be placed in test chamber and the temperature elevated gradually to 150°F and allowed to remain at this temperature until thermo-stabilization is reached within the container. The test sample shall remain at the thermo-stabilized temperature of 150°F for a period of 4 hours, after which the temperature shall be gradually reduced to ambient. There shall be no evidence of cracks, delamination or distortion as a result of the test. There shall be no leakage when the container is tested in accordance with 4.6.1.4.

(b) Low temperature.- The container assembly shall be placed in test chamber and the temperature lowered gradually to -65°F and allowed to remain at this temperature until thermo-stabilization is reached with-in the container. The test sample shall remain at the thermo-stabilized temperature of -65°F for a period of 4 hours, after which the temperature shall be gradually raised to ambient. There shall be no evidence of cracks, delamination or distortion as a result of the test. There shall be no leakage when the container is tested in accordance with 4.6.1.4.

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4.6.2.2 Cover and body.- The cover shall mate with body as specified in 3.7.2 when subjected to a functional examination by opening and closing the container at least five times and subsequently visually examining the components for defects or poor workmanship.

4.6.2.3 Cushion inserts.- A visual and tactile examination shall be performed to determine that the cushions are securely bonded in position. Visual examination shall reveal no evidence of damage or shifting of cushions.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- Unless otherwise specified, the container assembly shall be prepared for shipment using normal commercial methods, or materials to assure acceptance by common or other carrier for safe transportation at the lowest rate to point of destination.

5.2 Marking.- Marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The reusable container defined herein is intended for shipment and storage of the Laser Range Finder AN/GVS-3 (10540501) for issue as a replacement assembly.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Container assembly part number
(c) Order of document precedence
(d) Applicable stock number
(e) Requirements for first article tests

Custodian:

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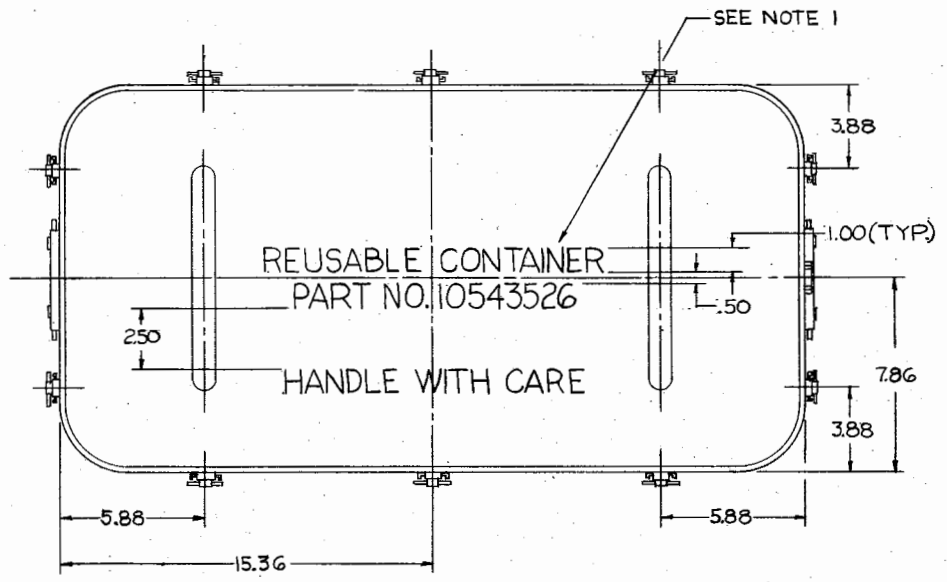
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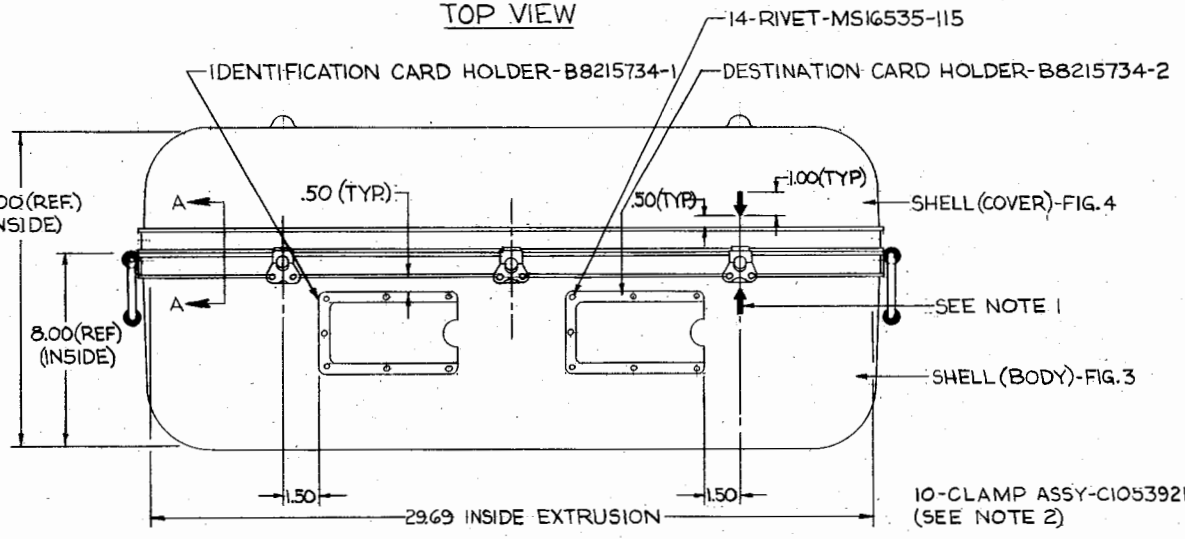
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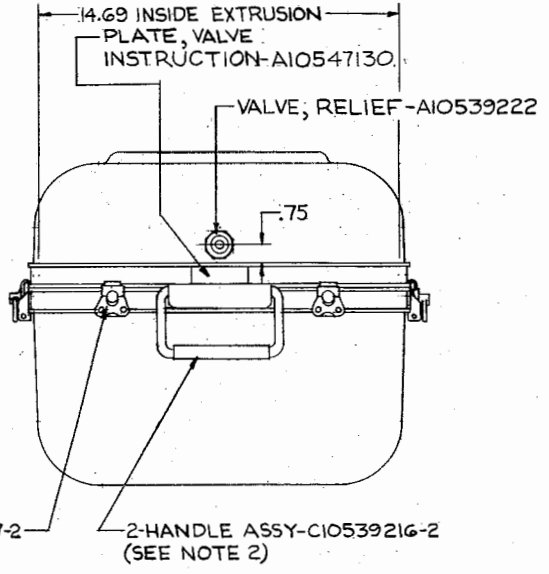
NOTES:
1. LEGEND SHALL BE APPLIED BY SILK SCREEN OR STENCIL AS SPECIFIED IN PAR. 3.6. CHARACTERS SHALL BE UPPER CASE VERTICAL GOTHIC WITHOUT SERIFS.
2. HARDWARE SHALL BE PERMANENTLY ATTACHED TO EDGE FRAME ASSEMBLY AS SHOWN ON FIG. 2.
3. TOLERANCE ±1/16



TOP VIEW



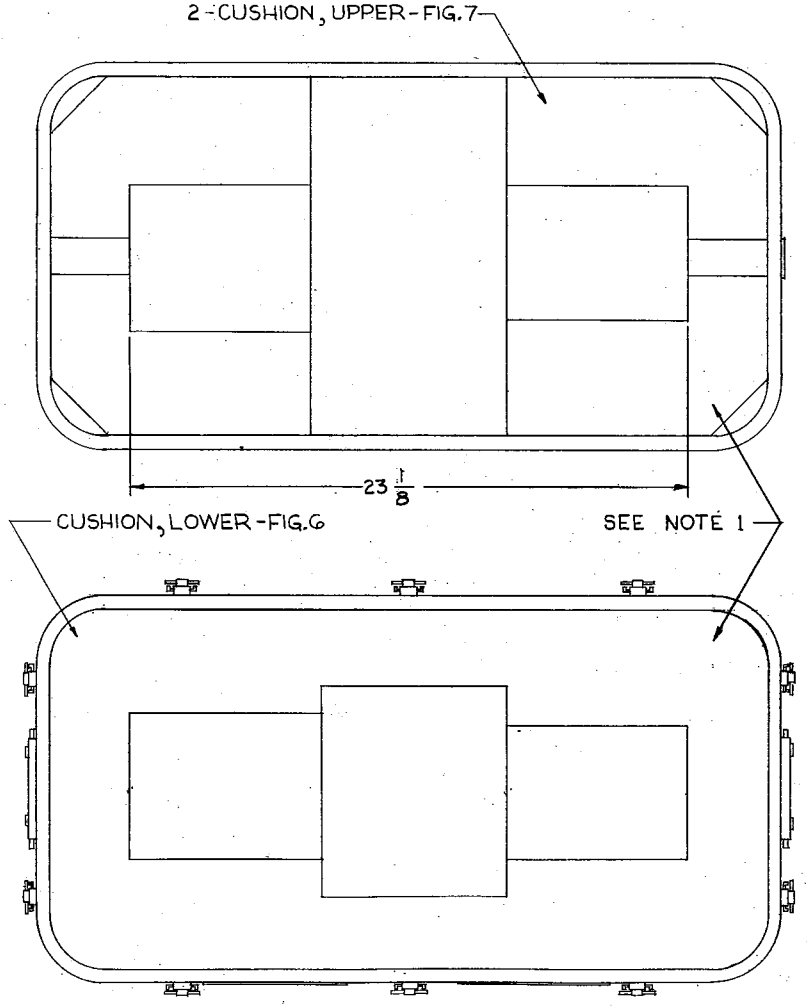
FRONT VIEW



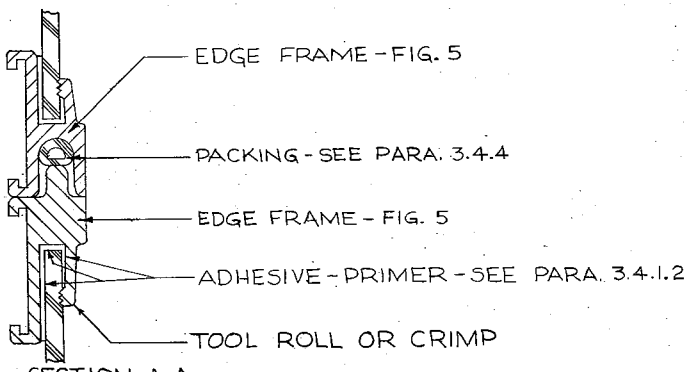
END VIEW

FIGURE 1-CONTAINER ASSY

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NOTE:
1. CUSHIONS SHALL BE BONDED IN POSITION WITH ADHESIVE CONFORMING TO MIL-A-5092, TYPE II.



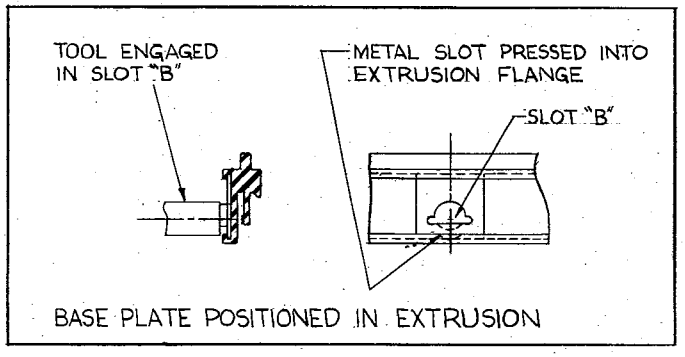
SECTION A-A

FIGURE 1A - CUSHION ARRANGEMENT

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INSTALLATION PROCEDURE:

1. LOCATE CLAMP, CLASP, HANDLE, & HINGE ASSEMBLIES, AS REQUIRED, IN POSITION AS SHOWN IN FIGURE 1.
2. POSITION HARDWARE BASE PLATE IN EXTRUSION FLANGE.
3. ENGAGE INSTALLATION TOOL IN SLOT "B", AS SHOWN.
4. FIRMLY PRESS TOOL DOWNWARD & TURN TOOL 90° TO THE RIGHT & 90° TO THE LEFT, TO DEFORM SLOT OF BASE PLATE IN EXTRUSION FLANGE AS SHOWN.



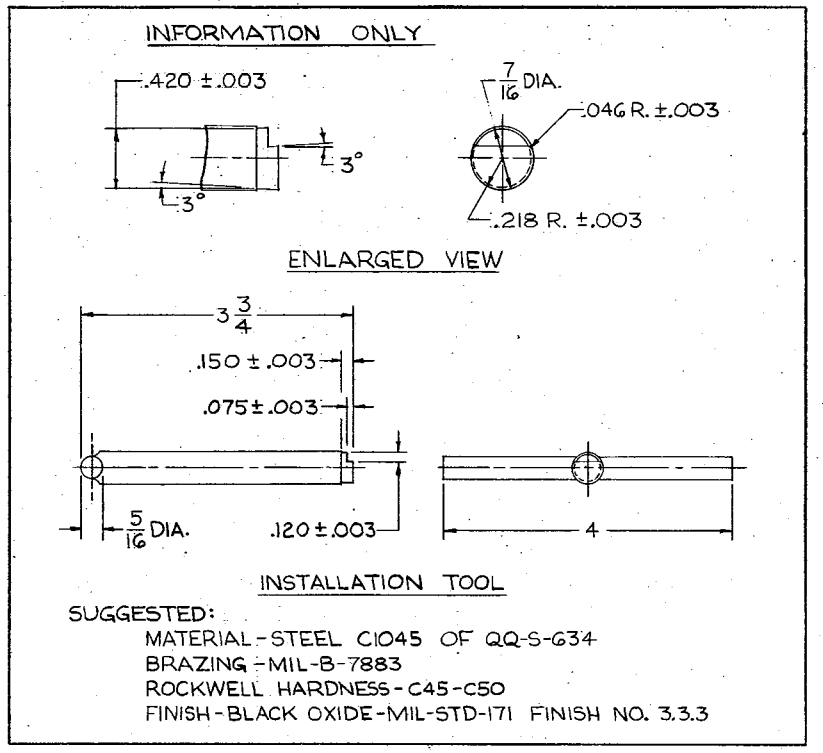
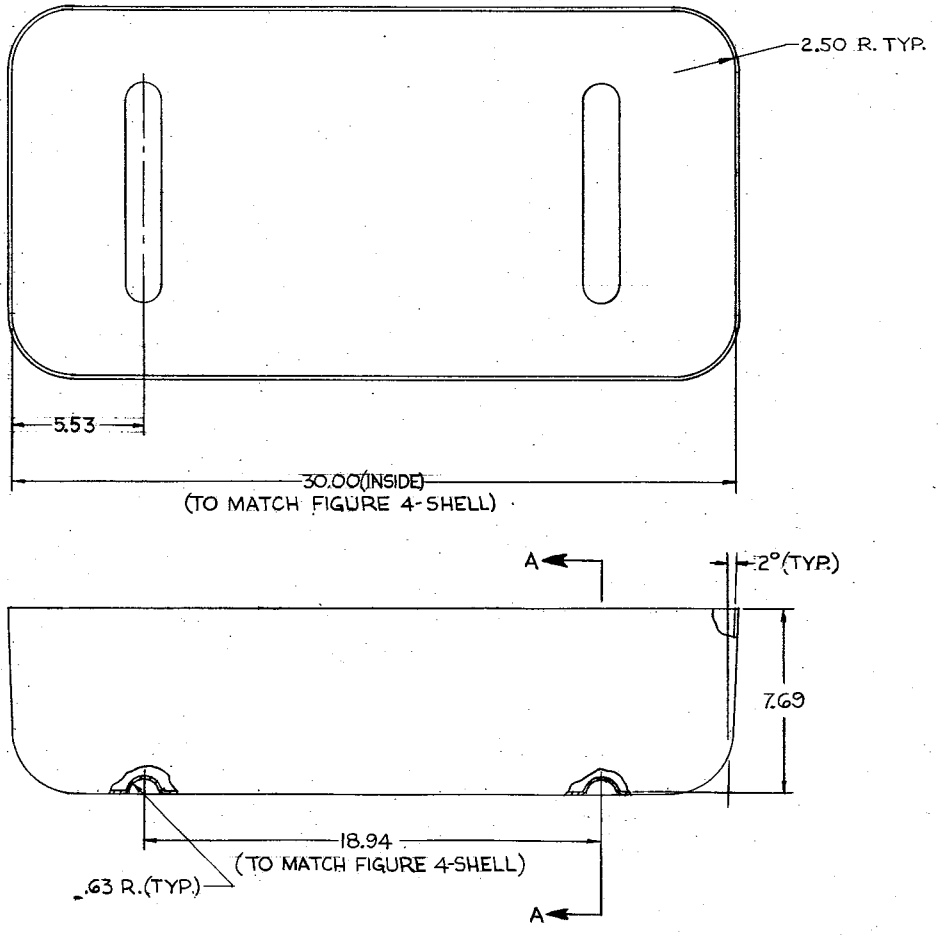


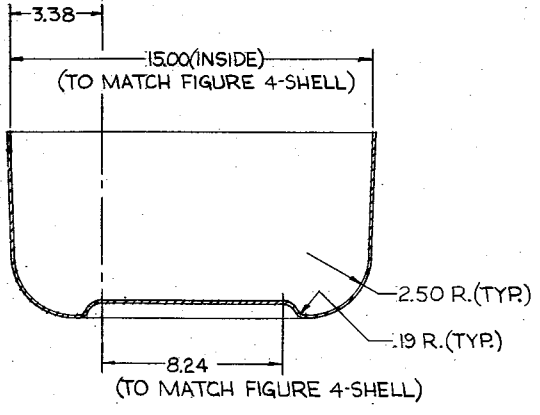
FIGURE 2 - INSTALLATION OF HARDWARE

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NOTES:
1. MATERIAL SHALL BE AS SPECIFIED ON A10541035, FORMED WALL THICKNESS .120 ±.020

COLOR-OLIVE DRAB, X24087 OF FED-STD-595
FINISH-HAIRCELL GRAIN (EXTERIOR), HAIRCELL GRAIN OR SMOOTH (INTERIOR-OPTIONAL)
2. TOLERANCE: DECIMALS ±.03; ANGLES +1°,-0°





SECTION A-A

FIGURE 3 - SHELL (BODY)

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NOTES:
1. MATERIAL SHALL BE AS SPECIFIED ON A10541035, FORMED WALL THICKNESS .120 ±.020

   COLOR-OLIVE DRAB, X24087 OF FED-STD-595
   FINISH-HAIRCELL GRAIN (EXTERIOR), HAIRCELL GRAIN OR SMOOTH (INTERIOR-OPTIONAL)
2. TOLERANCE- DECIMALS ±.03; ANGLES +1°, -0°

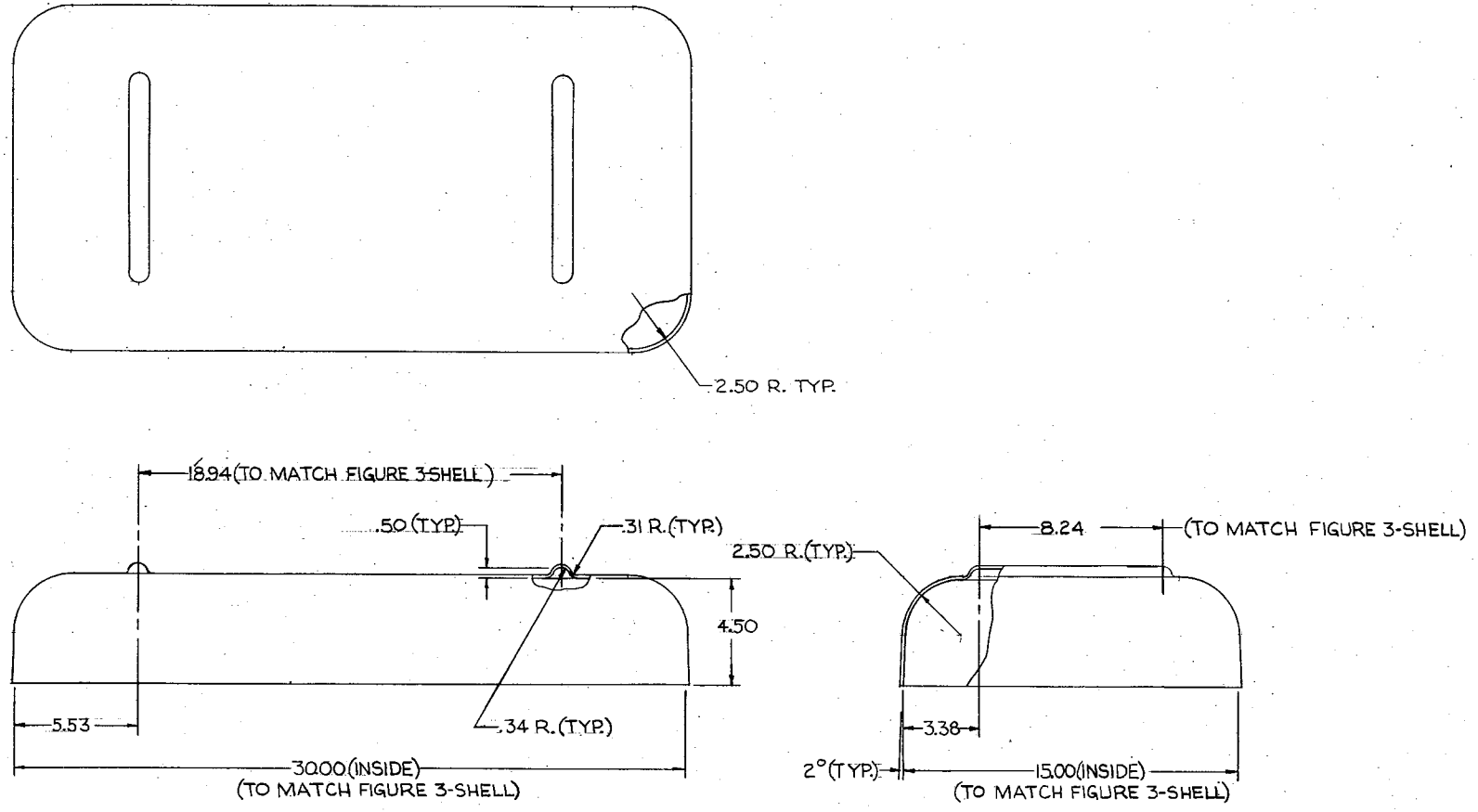
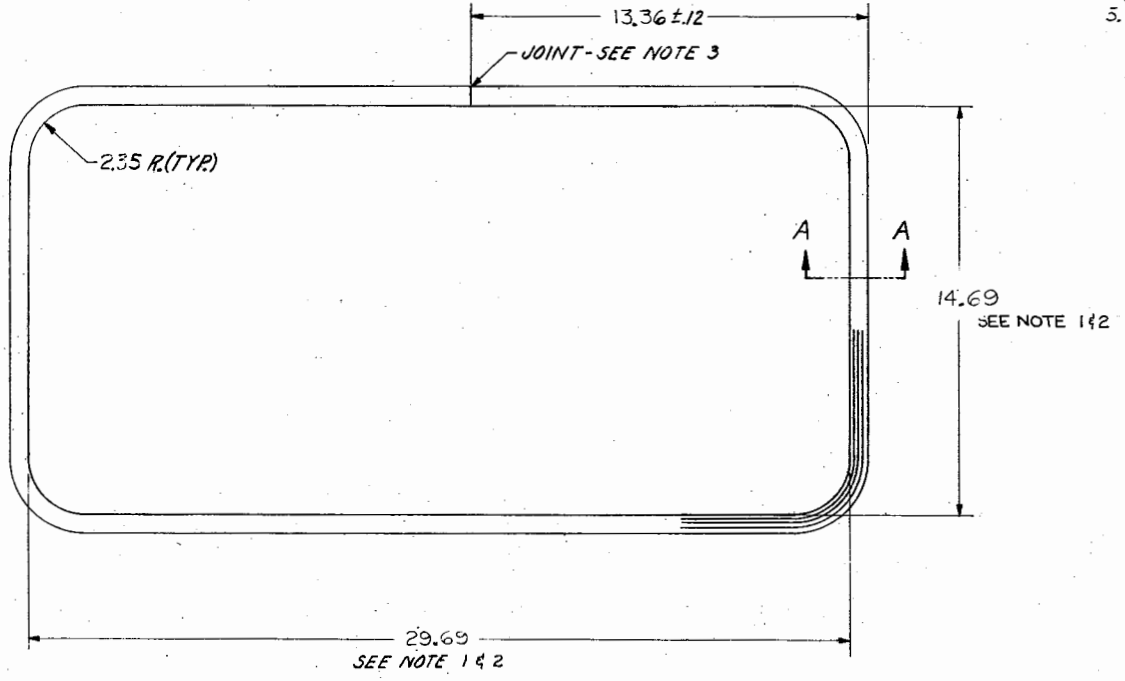
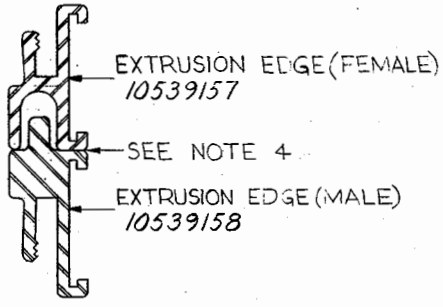


FIGURE 4-SHELL (COVER)

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NOTES:
1. COMPLETED ASSEMBLIES SHALL FIT FREELY INTO EACH OTHER AS SHOWN IN SECTION "A-A"
2. MUST MATCH SHELLS SHOWN ON FIGURES 3 AND 4
3. EXTRUSIONS SHALL BE FORMED, WELDED OR BRAZED, AND FINISHED IN ACCORDANCE WITH PARA. 3.4.1 AND 3.4.1.1
4. SURFACE SHOWN SHALL BE FLAT WITHIN .020 T.I.R. IN ANY 6 INCHES OF LENGTH AND .030 T.I.R. WITHIN ENTIRE LENGTH OF ANY SIDE OR END.
5. TOLERANCE ±.03



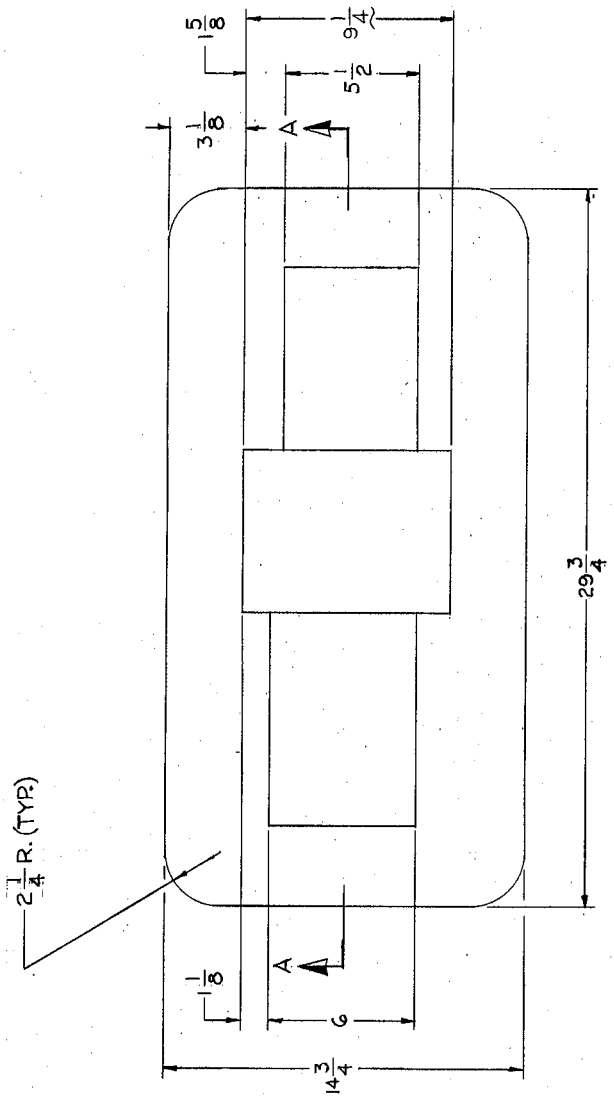


SECTION A-A

FIGURE 5-EDGE FRAME ASSEMBLIES

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NOTES:
1. CUSHION SHALL BE FABRICATED FROM POLYURETHANE-POLYETHER FORMULATION, CONFORMING TO MIL-P-26514, TYPE I, CLASS 2, COLOR-CHARCOAL, 1.4 LB./CU. FT. DENSITY
2. TOLERANCE ± 1/16



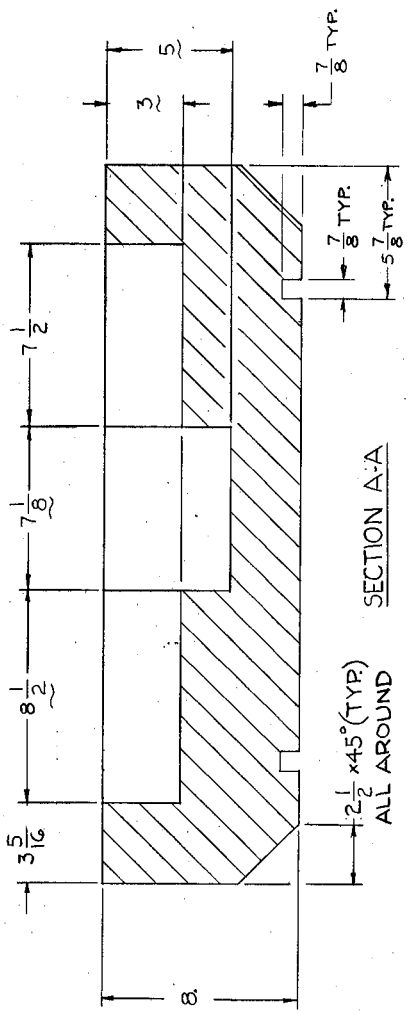
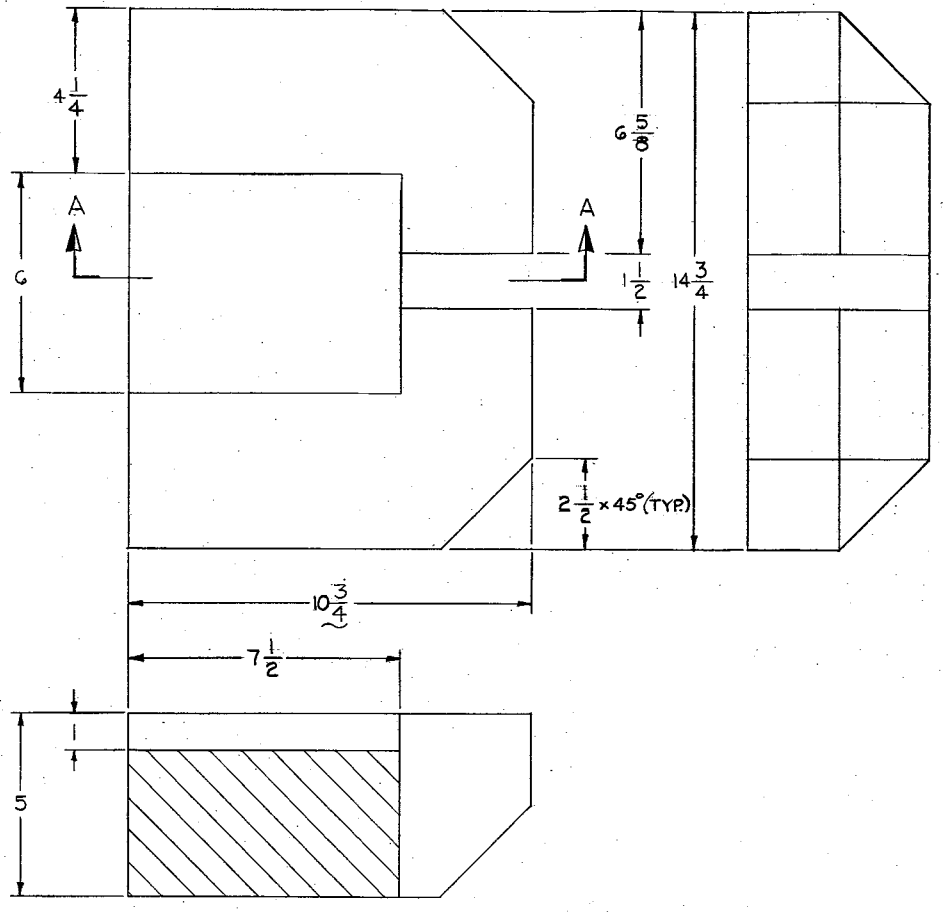


FIGURE 6-CUSHION, LOWER

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SECTION A-A

NOTES:
1. CUSHION SHALL BE FABRICATED FROM POLYURETHANE-POLYETHER FORMULATION, CONFORMING TO MIL-P-26514, TYPE I, CLASS 2; COLOR-CHARCOAL; 1.4 LB./CU.FT. DENSITY.
2. TOLERANCE ± 1/16

FIGURE 7-CUSHION, UPPER

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NOTES
1. MATERIAL-OPTIONAL
2. TOTAL WEIGHT- 20LBS.

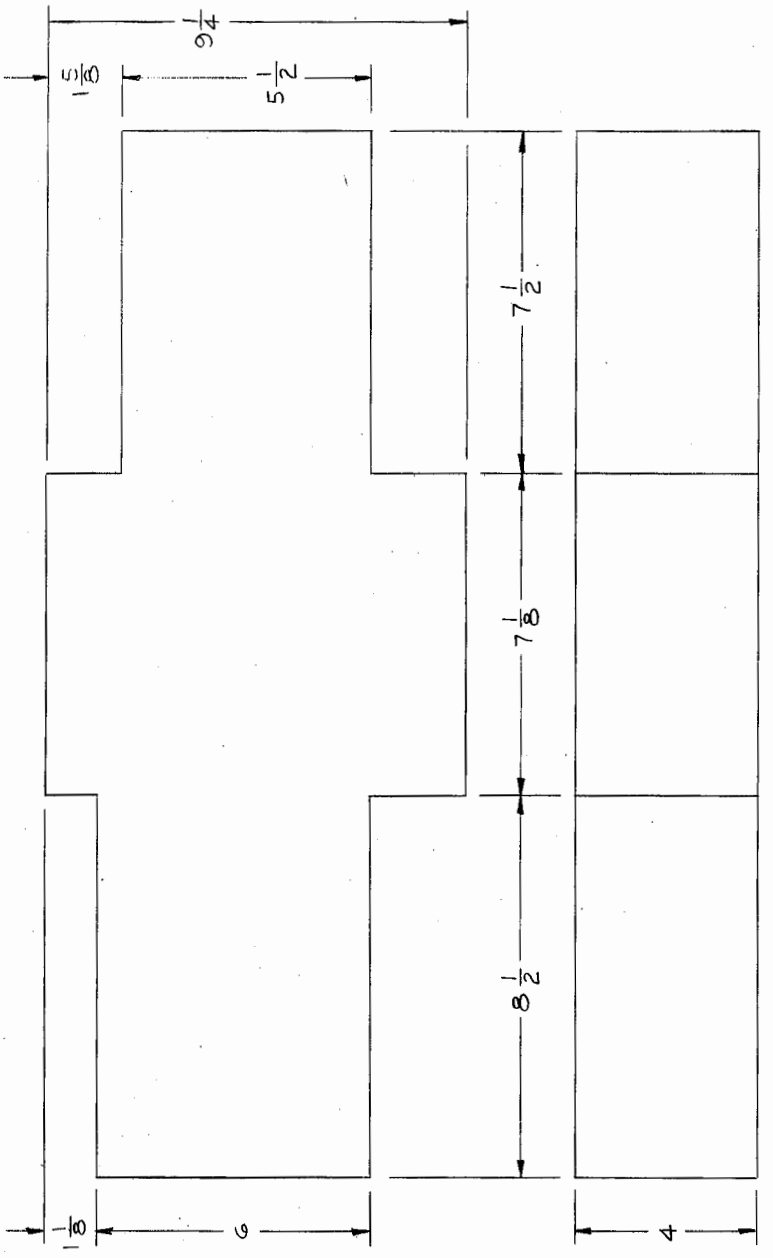


FIG. 8 - DUMMY LOAD

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